

JavaScript Lab Examination

Topic: Manipulating HTML with JavaScript (DOM)

Time: 1 hr 45 min

General Instructions

1. You are given a starter file `lab_student.html`. Open it in a text editor and in a web browser side by side.
2. **Only edit the `// TODO` sections inside the `<script>` tag.** Do not change the HTML structure or the `id` names of the elements.
3. After each change, **save the file and refresh the browser** to test your work.
4. You may only use what we have covered: `document.getElementById()`, `.innerHTML`, `.style`, `prompt()`, variables, operators, `if/else`, loops, strings (and string methods), and functions. Do not use arrays or objects.
5. Rename your final file to `<yourID>_lab.html` before submitting.

Quick Reminder — DOM Manipulation Technique

Every task uses the **same three steps**: a button click runs a function, the function finds an element, then changes its content or its style.

Goal	Code
Find an element by its id	<code>let el = document.getElementById("countOutput");</code>
Change its text/content	<code>el.innerHTML = "Hello";</code>
Change its colour / style	<code>el.style.color = "red"; el.style.backgroundColor = "yellow";</code>
Run code on a click	<code><button onclick="myFunction()">Click</button></code>
Ask the user for input	<code>let x = prompt("Enter a number:");</code>

Worked example — clicking the button changes the text and its colour:

```
<p id="msg">Hello</p>
<button onclick="change()">Click me</button>

<script>
```

```
function change() {  
  let p = document.getElementById("msg");    // 1. find  
  p.innerHTML = "You clicked it!";          // 2. change content  
  p.style.color = "green";                  // 3. change style  
}  
</script>
```

Task 1 — Click Counter

A number on the page starts at **0**. Make the three buttons work.

Requirements:

1. Complete `increaseCounter()` so that every click of the **Increase** button adds 1 to the variable `count` and shows the new value inside the `<span id="countOutput">`.
2. Complete `decreaseCounter()` so the **Decrease** button subtracts 1 from `count` and updates the display — but the counter must **not go below 0**.
3. Complete `resetCounter()` so the **Reset** button sets `count` back to 0 and updates the display.
4. **Bonus:** when the count reaches **10 or more**, change the number's colour to red using `.style.color` (and back to black after a reset).

Expected: Increase six times shows `6`; then Decrease twice shows `4`; Reset shows `0` again.

Task 2 — Text Case Converter

When the button is clicked, ask the user for a word or name and show it in different letter cases *on the page*.

Requirements:

1. Inside `convertCase()`, read the text from `prompt()`.
2. Build and display all of these in `<p id="textOutput">`, each on its own line:
 - the text in **UPPERCASE**
 - the text in **lowercase**
 - the text **Capitalized** — first letter uppercase, the rest lowercase
 - the **Length** (number of characters) of the text
3. **Bonus:** if the text is longer than 5 characters, colour the output blue, otherwise green, using `.style.color`.

Example: input `jaVaScRipt` → `JAVASCRIPT` / `javascript` / `Javascript` / Length: 10.

Task 3 — Traffic Light Simulator

A grey circle on the page is a traffic light. Each click of **Next Light** must move it to the next colour and show the matching instruction.

Requirements:

- 1. Inside `nextLight()` , increase the counter `lightStep` by 1 on every click.
- 2. Use `lightStep % 3` with an `if / else-if / else` chain so the light cycles in this order:

Step (% 3)	Light colour	Message in #lightOutput
1	red	STOP
2	yellow / orange	GET READY
0	green	GO

- 3. Change the circle using `document.getElementById("light").style.backgroundColor` , and write the message into `<p id="lightOutput">`.
- 4. **Bonus:** add a glow that matches the colour using `.style.boxShadow = "0 0 20px " + color;` and set the message text to the same colour.

Expected: repeated clicks cycle Red → Yellow → Green → Red ..., with the instruction text and (bonus) the glow changing to match each light.